

MICHELE GRIMALDI

SOFTWARE ENGINEER

CONTACT

Bari, Italy
EU Citizenship
Ready to relocate
Email: grimaldimichele806@gmail.com
Phone: +39 351 976 1913
GitHub: github.com/Mike014
LinkedIn: <https://www.linkedin.com/in/michele-grimaldi-dev/>

CORE SKILLS

LANGUAGES

C#, Python, Rust, C++, Java, Javascript

ARCHITECTURE

Event-Driven, CQRS, Event Sourcing,
Multithreading, Memory Models, Systems
Architecture

BACKEND

ASP.NET Core, Flask, REST API, PostgreSQL,
SQL

FRONTEND

React, Web development tools (DevTools)

PRACTICES

Unit Testing, Git, Agile, CI/CD pipelines,
Technical Documentation

INFRASTRUCTURE & CLOUD

Docker, GitHub Actions, Linux CLI, RunPod

TOOLS

Visual Studio, IntelliJ IDEA, Cursor, Xcode,
Android Studio, Slack

AI/ML

PyTorch, Keras, TensorFlow, Prompt
Engineering, Jupyter notebooks, Google Colab,
CUDA

LANGUAGES

Italian Native

English C1 Spanish A2

SUMMARY

I design and build software across every level of abstraction — from low-level systems to intelligent platforms. I choose the right tool for each problem — whether that means low-level memory management in C++, event-driven architecture in C#, ownership-based systems in Rust, or AI pipelines in Python. I approach every codebase the same way — **read the architecture first, understand the constraints, then deliver**. I thrive in unfamiliar codebases — reading architecture, tracing bugs, and delivering solutions whether it's a personal project, an open-source library with 30k+ commits, or a production enterprise system. **Looking to grow in a tech company that values engineering quality and continuous learning.**

WORK EXPERIENCE (2022 - Present)

Software Engineer — Neuram.ai · Contract May 2026 — Present

- Designed and deployed AI voice-agent pipelines (STT → LLM → TTS) in Python, managing cloud GPU infrastructure on RunPod.
- Deployed Voxtral-Mini-4B-Realtime-2602 via vLLM on RTX 4090 infrastructure, resolving CUDA compatibility issues and delivering a production-ready endpoint within a single day.
- Benchmarked real-time STT performance across HTTP, WebSocket, and concurrent execution scenarios, providing latency and cost analysis that informed production architecture decisions.
- Reduced LLM operational costs by 50–77% through benchmarking across five providers, leading to the adoption of the selected production model as the production model.
- Optimized production prompts by 67–71%, lowering token consumption, improving response latency, and supporting deployment of Italian-language voice agents.
Stack: Python | Linux CLI | CUDA | Docker | RunPod | Cursor IDE | Mistral Voxtra | Multithreading

Software Engineer Expert — Mercor · Freelance Mar 2026 — May 2026

- Evaluated AI-generated Python and C++ code for correctness, performance, and algorithmic quality on behalf of leading AI labs.
- Conducted benchmark-driven assessments using HumanEval and MBPP, identifying logical hallucinations and providing structured model feedback.
Stack: Python | C++ | C#

Career Break — Full-Time Training & Portfolio Development Feb 2025 — Mar 2026

- Completed a full-time Software Development Master focused on C#.NET and modern software engineering practices.
- Built OMNIA, a deep-learning audio spectrogram autoencoder using TensorFlow and Keras, and initiated development of Nexus Engine, an event-driven trading engine based on CQRS and Event Sourcing.
- Continued IBM AI Engineering Specialization, contributed to open-source projects, and maintained active competitive programming practice through LeetCode and CodeForces.

Prompt Engineer — MINDTEL (Mistral AI) / Outlier AI · Freelance Nov 2024 — Feb 2025

- Designed and optimized prompts for LLM code generation and reasoning tasks across Mistral models (Mistral-7B, Mixtral).
- Evaluated and validated AI-generated Python solutions, identifying logical errors and improving algorithmic correctness.
Stack: Python | Prompt Engineering

Software Engineer — Chordboard LLC · Freelance Sep 2024 — Jan 2025

- Led cross-platform porting of a professional audio production application from iPadOS to Windows and iOS using C++ and JUCE, delivering the transition in 5 months.
- Designed a unified preprocessor-based build architecture with automatic OS detection, reducing platform-specific code branches from 12 to 3 and unifying 70% of the codebase.
- Conducted Android feasibility analysis, achieving the first successful application execution on Android and identifying architectural blockers that informed product roadmap decisions.
- Managed the complete Apple deployment pipeline (certificates, provisioning profiles, App Store APIs) and performed cross-platform validation across iPadOS, iOS, and Windows.
Stack: C++ | JUCE | CMake | Xcode | Android Studio | Slack | Git

Audio Programmer & Software Developer — Freelance Jan 2022 — Aug 2024

- Combined advanced audio software training (with MasAcoustics) with hands-on development work across commercial, educational, and freelance projects.
- Developed real-time audio software and DSP tools using C++, JUCE, Csound, and Max/MSP.
- Designed and implemented low-latency audio processing pipelines, gaining practical experience with multithreading, memory management, and real-time systems constraints.
- Built custom audio applications, plugins, and prototyping tools spanning desktop software, sound design, and generative audio environments.
Stack: C++ | JUCE | Csound | Max/MSP | DSP | Multithreading | Audio Programming | Real-Time Systems | Git

ADDITIONAL EXPERIENCES & PROJECTS

• Top 1% Python Engineer globally (Algora Index, 2026). Active open-source contributor with verified merged pull requests.

• Competitive Programming — LeetCode & CodeForces. Algorithmic problem solving across Python, C++, Java, and C# depending on performance and implementation constraints.

• Language Progression (2015–Present) — VB6 → Csound → Max/MSP → C → C++ → C# → Python. Multi-paradigm evolution spanning DSP, systems programming, enterprise software, and AI engineering.

• Java Utilities (Ongoing) — Dependency-free reusable libraries with strict JUnit 5 test coverage and focus on clean architecture principles.

• Game Jams (2023–Present) — Participated as programmer and audio developer, delivering playable projects under extreme time and scope constraints.

• Computer Science Foundations (2018) After completing an IT Accounting diploma, started a Computer Science degree before pivoting to audio technology, sound design, and audio software development.

EXTRA

LeetCode: Mike014
YouTube: @AuTomA_014

KEY PROJECTS

Horizon — Compiled Programming Language Jun 2026 – Present
Statically typed, compiled language with native code generation, designed for high-performance backends, games, and AI tooling (Rust)
> Designed the full language specification and memory model. Developed in Agile phases — Phases 0–3 complete (lexer, parser, checker, generics, traits, modules, native binaries), Phases 4–5 in progress. Compiler (Rust/Cranelift) **built with Fable (Anthropic)**.
Stack: Rust | Cranelift | LLVM (planned) | FFI C | AI-Assisted Development (Fable, Anthropic)

Sentinel — AI-Powered Anomaly Detection Engine Jun 2026 – Present
Real-time system metrics anomaly detection engine (Rust + Keras/Python)
> Rust collector with thread-safe circular buffer feeds an LSTM Autoencoder via TCP socket IPC. TFLite FP16 inference at 5ms latency. Detection engine: F1 0.86, AUC-ROC 0.997, 99th-percentile dynamic threshold.
Stack: Rust | Python | Keras | TFLite | TCP Socket IPC | LSTM Autoencoder | scikit-learn

Nexus Engine — Simulated Order Book Exchange Engine Released June 2026
A simulated order/betting engine (C#)
> Built across 5 Agile sprints demonstrating mastery of transactional systems, concurrency, real-time communication, and event-driven architecture.
Stack: C# | ASP.NET Core 9 | PostgreSQL 16 | MediatR | SignalR | React | TypeScript | Event Sourcing | CQRS | Optimistic Locking | Docker | Railway | Vercel

NanoStore Completed June 2026
Nanostore is a C++23 embedded key-value storage engine designed for crash resilience and thread safety (C++)
> Binary WAL, CRC32 integrity, atomic compaction, std::shared_mutex concurrency. 22 test cases, 1537 assertions.
Stack: C++ | C++23 | Catch2

NetProbe Completed June 2026
TCP Latency Benchmark Tool (C++)
> Real-time RTT measurement using CPU cycle-accurate timing (RDTSC), color-coded spike detection, cross-platform Winsock2/POSIX architecture containerized via Docker Compose for zero-install execution.
Stack: C++ | CMake | Winsock2 | POSIX Sockets | Docker

DEAD AIR Released May 2026
Unity Video Game (C#)
> Unity horror video game with event-driven C# architecture: tag-driven framework using Observer Pattern across 14 event channels, <1ms CPU/frame and ~2MB RAM at idle.
Stack: C# | .NET | Observer Pattern | Dependency Injection | ScriptableObjects

OMNIA Completed Mar 2025
Audio Spectrogram Autoencoder (Python)
> Deep Convolutional Autoencoder that reconstructs and denoises audio spectrograms using CNNs trained on the ESC-50 dataset.
Stack: Python | Keras | TensorFlow | Librosa | ESC-50

Remote Jobs App Completed Jan 2025
Web application for exploring remote-friendly companies, deployed on Railway/Render(Python)
> Full-stack Python web app for remote job search: Flask + SQLAlchemy backend, Bootstrap UI, deployed in production on Railway.
Stack: Python | Flask | SQLAlchemy | Bootstrap | Railway/Render

OPEN SOURCE CONTRIBUTIONS (2024 - Present)

OpenRA (C# RTS Engine)
LANGUAGE: C# - Traced and resolved bugs across C#, YAML, and Fluent on a 30k+ commit codebase -- auto-targeting bypass analysis (#22456, informed merged fix #22471), generated map error handling (#22260), Map Editor localisation (#22241). Codebase-first methodology, all PRs passed CI with maintainer review.

Microsoft PowerToys (Windows Utilities)
LANGUAGE: C# - Opened PR #47768 on Advanced Paste, tracing full MVVM data flow across 8 files and 7 architectural layers. PR passed CI. Maintainers redirected toward a new feature — currently in development.

NLTK (NLP Toolkit)
LANGUAGE: Python - Resolved legacy CoreNLPParser API issue (#2112). Property overloading for input data integrity. Merge approved by project maintainer.

Marten (.NET Event Sourcing & Document DB)
LANGUAGE: C# - Reported Issue #4443 on lock pattern optimization (object → System.Threading.Lock). Issue validated and resolved by lead maintainer (PR #4448).

Google Guava (Java Core Libraries)
LANGUAGE: Java - Documented missing thread-safety contract on Closer class (issue #3900), applied to both JRE and Android flavors. proposed Javadoc fix. PR #8470, CI passed.

DocsGPT (Open Source RAG Platform)
LANGUAGE: Python - Resolved model/→models/ folder consolidation (#1445, PR #1603). Traced all references across Dockerfile, vectorstore, and install scripts via grep analysis. Tested full lifecycle. PR merged by maintainers.

EDUCATION & CERTIFICATIONS

• **Princeton University (Coursera) — Computer Architecture**

Assembly, ISA, CPU, Cache, Parallelism (2026 – Present)

• **IBM — AI Engineering Professional Certificate**

Keras, PyTorch, Jupyter, Google Colab, Deep Learning, Neural Networks, Machine Learning Pipelines, LangChain, RAG, LLM Fine-Tuning Pipelines (2024 – 2026) Verify: coursera.org/verify/professional-cert/MNSWUYRT1ORT

• **Ongoing Professional Development**

- **Anthropic Academy:** Claude API, MCP, Claude Code
- **Microsoft Learn:** ASP.NET Core, C#.NET
- **NVIDIA Deep Learning Institute:** CUDA Fundamentals
- **edX:** Advanced Software Engineering Topics

• **Board Infinity (Coursera) — Java Multithreading Mastery**

Advanced Java Concurrency & Multithreading (Jun 2026)

• **ManPower / Challenge Network — Full Stack Web Developer**

Java, JavaScript, React, Browser Fundamentals (May 2026 – Jul 2026)

• **Epicode Institute of Technology — Master in Software Development**

C#, .NET, Git, Deployment Services (Sep 2025 – May 2026)

• **Codecademy — Professional Certification in Computer Science**

Python, Algorithms & Data Structures — Grade: 86/100 (Jan 2024 – Jan 2025)

• **I.T.E.T. Cassandro Fermi Nervi — High School Diploma in IT Accounting**

Visual Basic, Business Information Systems & Computer Science Foundations (Sep 2015 – Jul 2018)

COMPETITIONS

Kaggle Competition Participant — AI Agent Security | OpenAI / Google / IEEE Jun 2026 – Present

- **Developed automated vulnerability discovery algorithms** for tool-using LLM agents within a deterministic security testing environment.

Stack: Python | Go-Explore

Ready Tensor Agentic AI Innovation Challenge | NoVa (Autonomous AI Agent) Feb 2025 – Apr 2025

- **Developed NoVa**, an **autonomous agent** featuring **dynamic multi-model inference** (Hugging Face), **persistent SQLite state**, and **real-time web navigation via WebVoyager** implementation using open-weight LLMs.

Stack: Python | Hugging Face | SQLite | WebVoyager